

1 / 5 | Ages 7-8 | 10 minutes

Three pets, two clues

What we are practising

Combine clues to find a unique one-to-one match.

What you need

Paper / Pencil

Use spoken names and point to drawn animals.

Before you begin

Write Sami, Noor and Huda; each has one different pet: cat, bird or fish.

All characters are fictional; no pets are needed for this activity.

Try it together

1. Draw a grid with the names as rows and animals as columns.
2. Use these clues: Sami has the fish; Noor does not have the bird.
3. Cross out impossible matches and fill the remaining ones.
4. Check that every animal and every child appears exactly once.

Talk, draw or show your idea

Which clue should you use first, and why?

Space to explore

2 / 5 | Ages 7-8 | 15 minutes

Two sorting rules

What we are practising

Classify items that meet both, one or neither of two rules.

What you need

Paper / Two coloured pencils

Write R or B inside shapes instead of using colours.

Before you begin

Draw two overlapping circles labelled red and triangle.

Use labels as well as colours so colour recognition is not required.

Try it together

1. Draw a red triangle, blue triangle, red square and blue square beside the circles.
2. Copy each shape into the place matching its properties.
3. Explain why one shape belongs in the overlap.
4. Invent a new shape for each region, including outside both circles.

Talk, draw or show your idea

Where does a blue square belong?

Space to explore

3 / 5 | Ages 7-8 | 15 minutes

Robot on a grid

What we are practising

Sequence commands and locate the first instruction that causes an error.

What you need

Paper / Pencil / Large paper marker

Move a fingertip instead of a marker.

Before you begin

Draw a 4 by 4 grid; mark start at bottom left, goal at top right and a blocked square directly above start.

Keep the robot on paper; no blindfolded walking is needed.

Try it together

1. Agree that each command moves one square up, down, left or right on the page.
2. Write commands to reach the goal without entering the blocked square.
3. Ask a partner to follow the commands literally.
4. If it fails, change the first incorrect command and test again.

Talk, draw or show your idea

What changes if the first command is up?

Space to explore

4 / 5 | Ages 7-8 | 10 minutes

Find a rule, test a rule

What we are practising

Apply a consistent rule while recognizing that a short pattern may have several explanations.

What you need

Paper / Pencil

Say the sequence aloud and count with fingers.

Before you begin

Write 2, 5, 8, 11 and leave two blanks.

Allow thinking time and accept explained alternative rules.

Try it together

1. Compare each pair of neighbours.
2. Choose one rule that fits every shown step.
3. Use that rule to fill the blanks and check backward.
4. Create another four-number pattern for a partner.

Talk, draw or show your idea

What evidence supports your rule?

Space to explore

5 / 5 | Ages 7-8 | 15 minutes

The snack-box order

What we are practising

Combine first, last and immediately-before relationships.

What you need

Paper / Pencil

Use four large folded paper cards.

Before you begin

Label four drawn boxes apple, bread, carrot and dates; positions run left to right.

Use drawn food; no tasting or food preparation is involved.

Try it together

1. Draw four numbered positions from 1 to 4.
2. Use the clues: apple is first; dates is last; bread is immediately before carrot.
3. Place the boxes and read every clue again.
4. Explain why swapping the middle two boxes fails.

Talk, draw or show your idea

Does before always mean immediately before?

Space to explore

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

1. Three pets, two clues

A little hint

Once Sami has the fish, only cat and bird remain for the others.

Answer / what to notice

Sami has the fish, Noor has the cat and Huda has the bird.

Noor cannot have the used fish or the forbidden bird, so she has the cat; Huda gets the remaining bird.

Make it easier

Remove Sami and solve only the cat-or-bird choice.

Take it further

Write a new set of clues with a different unique answer.

Another day, away from the screen

Create a matching puzzle about three bags and three colours.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

2. Two sorting rules

A little hint

The overlap means red AND triangular.

Answer / what to notice

Red triangle: overlap. Blue triangle: triangle only. Red square: red only. Blue square: outside both.

An item belongs in a circle only when it meets that circle's rule.

Make it easier

Start with separate groups for red and blue.

Take it further

Replace colour with size and define a clear size cutoff.

Another day, away from the screen

Sort imagined animals by has wings and lives mainly in water.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

3. Robot on a grid

A little hint

Move right before moving up to avoid the blocked square.

Answer / what to notice

One shortest route is right, right, right, up, up, up; it uses six moves.

The goal is three columns right and three rows up, so fewer than six moves cannot reach it.

Make it easier

Use a 3 by 3 grid without a blocked square.

Take it further

Add a second obstacle and find two different shortest routes.

Another day, away from the screen

Give a pretend robot instructions to arrange three drawn books.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

4. Find a rule, test a rule

A little hint

Subtract each number from the next one.

Answer / what to notice

Using add 3 gives 14 and 17. Other explicitly stated rules may also fit the short sequence.

Each shown gap is 3; stating the rule makes the prediction checkable without claiming it is the only possible rule.

Make it easier

Use 2, 4, 6, 8 and count pairs.

Take it further

Try a repeating rule such as add 2, then add 5.

Another day, away from the screen

Make a repeating pattern with claps and taps and state its rule.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

5. The snack-box order

A little hint

After filling the two ends, only positions 2 and 3 remain.

Answer / what to notice

From left to right: apple, bread, carrot, dates.

The fixed ends leave two spaces, and the immediate-before clue fixes their order.

Make it easier

Place the first and last boxes for the child.

Take it further

Remove the last-position clue and find every order still allowed.

Another day, away from the screen

Create a new four-item order and give a partner enough clues to solve it.